

MIDTERM EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A liquid film of constant thickness δ is flowing down an inclined flat conveyor belt of length L and width W that is moving with a constant velocity v_0 .

- 1- Develop a shell momentum balance to obtain the steady-state velocity profile in the liquid film. Assume L and W to be large compared to δ , so the disturbances at the edges of the system may be ignored.
- 2- Determine the viscous force of the liquid film on a unit area of the conveyor belt.
- 3- Determine the mass flow rate of the liquid film.

Your answers may be in terms of the following constant parameters:

μ and ρ : viscosity and density of the liquid

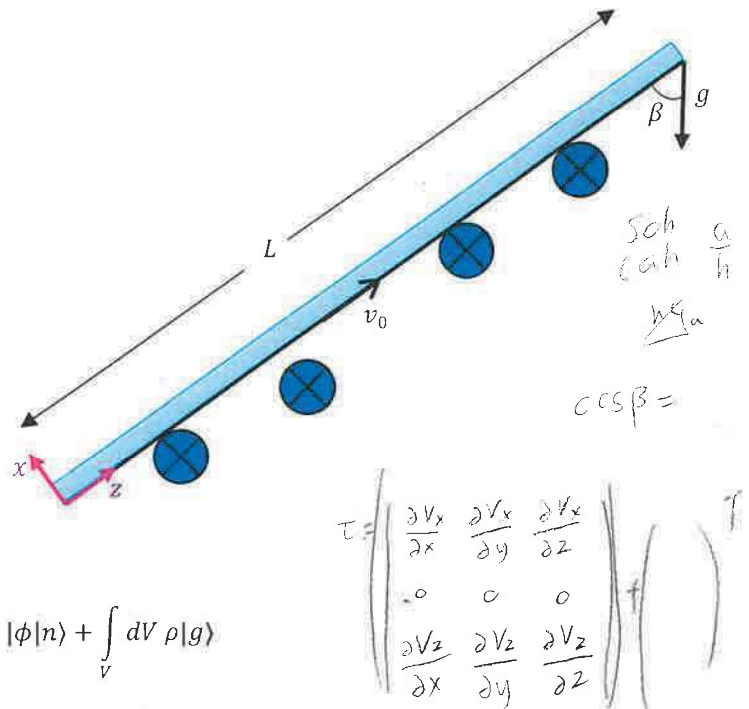
δ : thickness of the liquid film

L and W : length and width of the conveyor belt surface.

v_0 : velocity of the conveyor belt surface in z -direction.

g : gravitational constant.

β : angle between the conveyor belt surface and the direction of gravity.



$$\int_V dV \frac{\partial \rho(v)}{\partial t} = - \int_S dS (\phi \cdot n) + \int_V dV \rho(g)$$

$$|\phi| = \rho(v)(v) + |\pi| = \rho(v)(v) + |\tau| + p|\delta|$$

$$|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$$

$$\tau = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ 0 & 0 & 0 \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{pmatrix} + \begin{pmatrix} p \\ 0 \\ 0 \end{pmatrix}$$

$$0 = \delta W \rho v_z v_z|_{z=0} - \delta W \rho v_z v_z|_{z=L} + L W \tau_{zx}|_{x=0} - L W \tau_{zx}|_{x=\delta} + L W P - L W P + \delta L W \rho g$$

$$p = p - \rho g \delta \cos \beta$$

$$0 = \delta W \rho v_z v_z|_{z=0} - \delta W \rho v_z v_z|_{z=L} + L W \tau_{zx}|_{x=0} - L W \tau_{zx}|_{x=\delta} + \delta W (p_0 - p_L)$$

because constant velocity at top and bottom

$$0 = \frac{L W \tau_{zx}|_{x=0} - L W \tau_{zx}|_{x=\delta} + \delta W (p_0 - p_L)}{\delta L W}$$

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$$0 = \frac{\tau_{zx}|_{x=0} - \tau_{zx}|_{x=\delta}}{\delta} + \frac{(P_0 - P_L)}{L}$$

$$-\frac{(P_0 - P_L)}{L} = \frac{d\tau_{zx}}{d\delta}$$

$$\frac{(P_L - P_0)}{L} = \frac{d}{d\delta} \tau_{zx}$$

$$\tau_{zx} = -\mu \left(\frac{\delta V_z}{\delta x} + \frac{\delta V_x}{\delta z} \right)$$

no V_x

$$\tau_{zx} = -\mu \frac{\partial V_z}{\partial x}$$

$$\frac{(P_L - P_0)}{L} = -\mu \frac{d^2 V_z}{dx^2}$$

$$\frac{(P_0 - P_L)}{\mu L} = \frac{d^2 V_z}{dx^2}$$

$$\frac{d^2 V_z}{dx^2} = \frac{(P_0 - P_L)}{\mu L}$$

$$\frac{dV_z}{dx} = \frac{(P_0 - P_L)}{\mu L} x + C_1$$

$$V_z = \frac{(P_0 - P_L)}{2\mu L} x^2 + C_1 x + C_2$$

$$V_z = \frac{(P_0 - P_L)}{2\mu L} x^2$$

at $x=0$, $V_z=0$

$$0 = 0 + 0 + C_2$$

$$C_2 = 0$$

at $x=\delta$, $V_z=0$

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$$3. V_z = \frac{(P_0 - P_L)^2}{2\mu L} x$$

$$w = V_z \cdot \rho$$

$$w = \frac{\rho(P_0 - P_L)^2}{2\mu L} x^2$$
